Shreyas Patil

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TECHNICAL SKILLS

- **Languages:** Python, C++, SQL
- **Frameworks & Concepts:** PyTorch, TensorFlow, CUDA, FastAPI, WebSockets, Temporal, Ray, Multi-Agent Systems (MAS), RAG
- **Libraries:** NumPy, Pandas, Scikit-learn, OpenCV, Librosa, PyArrow, Pydantic, OpenTelemetry, AsyncIO, SpaCy, DSPy, LiteLLM
- **Databases:** PostgreSQL (Aurora), MongoDB, Redis, Kuzu DB, SQLite
- **Developer Tools:** Git, Docker, Kubernetes, ArgoCD, Grafana, Langfuse, Twilio, FHIR, AWS

EXPERIENCE

Data Scientist 1 | *Xcaliber Health* **Sept 2025 - March 2026**

- Designed and deployed **"Merlin"**, a production multi-agent system running on **Ray Serve** that orchestrates secure healthcare workflows. Built complex tool integrations enabling agents to autonomously manage appointment scheduling, prescription refills, and demographic updates via strictly-typed FHIR payloads to **Athena EHR**.
- Owned and shipped an **asynchronous document validation workflow** for demographic updates, allowing patients to upload IDs mid-call via **WhatsApp** or SMS magic links (**S3 presigned URLs**). Integrated the workflow with the **Clinical Document Processing (CDP)** pipeline and replaced 3-second polling loops with event-driven updates, instantly returning extracted document data to the live voice session after processing, eliminating dead-air latency and preventing dropped calls.
- Collaborated on migrating a monolithic voice agent to a **dynamic routing architecture** for seamless mid-call handoffs between specialized **AWS Bedrock** domain agents. Optimized the caller experience by resolving LLM barge-in hallucinations via explicit **Twilio** audio-buffer flushes and designing latency-masking audio cues to naturally hide 5+ second API latencies.
- Designed and implemented the initial iteration of a long-term memory module for the internal **Agent Development Kit (ADK)**, enabling agents to maintain continuity across sessions while minimizing context-window usage. Buffered active conversations in **Redis** while a **Temporal**-backed async pipeline distilled salient facts, entities and relationships at session end and persisted them into a **Kuzu DB** knowledge graph.
- Resolved two production-critical failures in the **ADK**'s agent runtime: a session-isolation defect that leaked conversational context across concurrent healthcare voice calls and a state-propagation bug that stalled multi-step LLM execution. Redesigned DAG state management and migrated session history to **Redis**, enabling safe concurrent execution and horizontal scaling across **Ray Serve** replicas.
- Accelerated a **Temporal**-backed **Clinical Document Processing (CDP)** pipeline's OCR processing by **70%** across **10,000+ pages** of medical records by benchmarking frontier vision models and parallelizing CPU-intensive image preprocessing using a dedicated process pool.
- Slashed LLM token consumption by over **60%** and minimized API costs by implementing document-specific entity extraction mappings, eliminating the indiscriminate processing of irrelevant medical data.

Machine Learning Intern | *Stealth Startup* **Feb 2025 - May 2025**

- Prototyped an **LLM-driven pipeline** for automated technical assessments, utilizing Python (`ast`) to extract candidate code features for dynamic analysis.
- Improved evaluation pipeline reliability by building **Docker** containers and **Kubernetes** deployment configurations for agentic workflows.

Open Science Contributor | *Cohere For AI (Expedition Aya)* **July - Sept 2024**

- Implemented data filtering and toxicity analysis on a massive, 8-language **image-text dataset** used to pretrain the **Maya** vision-language model.
- Conducted cross-lingual verification for cultural coherence, directly contributing to the project winning the "Most Promising Project" award at Expedition Aya 2024.

PROJECTS

Contextually and Culturally Aware MT across Indian Languages

- Developed an enhanced **Machine Translation** pipeline tailored for India's diverse linguistic landscape, building upon the **IndicTrans2** model as foundation.
- Designed and implemented crucial pipeline components for **context extraction** (NER, sentiment analysis) and **cultural adaptation** to ensure translation relevance.
- Applied **Low-Rank Adaptation (LoRA)** to achieve significant **inference optimization**, reducing computational requirements and enabling deployment on resource-constrained **edge devices**.
- Performed comprehensive evaluations, demonstrating improved translation quality via standard metrics (**BLEU**, **chrF++**) across multiple language pairs.

Aspect Based Sentiment Analysis using Attention Based Bi-LSTM

- Leveraged **Attention mechanisms** to focus the context window more effectively around specific aspects within sentences.
- Utilized Attention-based **Bi-directional LSTM (Bi-LSTM)** networks to capture richer contextual information by processing text in both forward and backward directions.
- Developed an enhanced model for the **ABSA** task, achieving improvements in both prediction accuracy and training efficiency.

PUBLICATIONS

Patil, S., et al. (2025). *Towards a Nuanced Sentiment Analysis: A Novel Hybrid Approach to Aspect-Based Insights.* 2025 1st IEEE Uttar Pradesh Section WIE International Conference (UPWIECON). IEEE.

Patil, S., et al. (2024). *Leveraging Deep Learning for Real-Time Multilingual Translation in India's Linguistic Landscape.* International Conference on Futuristic Technologies in Control Systems & Renewable Energy (ICFCR). Springer, Singapore.

EDUCATION

New Horizon College of Engineering (VTU) | *Bengaluru* **Nov 2021 – Jun 2025**

- **Degree:** B.E. in Artificial Intelligence and Machine Learning **CGPA: 9.8/10**
- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, DSA and OOP Concepts, Database Management Systems, Linear Algebra, Probability, Statistics